

Batch: A2 Roll No.: 16010324044
Experiment / assignment / tutorial No. 7
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Dynamic Routing protocol (RIP) using Cisco Hardware

AIM: To configure Dynamic Routing protocol (RIP) for given network topology using Cisco Packet Tracer.

OUTCOME: Student will be able to analyze various routing techniques and design complete network as per the given requirements

Theory:

Dynamic Routing describes the capability of a system, through which routes are characterized by their destination, to alter the path that the route takes through the system in response to a change in conditions. The adaptation is intended to allow as many routes as possible to remain valid (that is, have destinations that can be reached) in response to the change.

People using a transport system can display dynamic routing. For example Adaptive routing can be seen within financial markets. For example, ASOR or Adaptive Smart Order Router (developed by Quod Financial), takes routing decisions dynamically and based on real-time market events.

The term is commonly used in data networking to describe the capability of a network to 'route around' damage, such as loss of a node or a connection between nodes, so long as other path choices are available. There are several protocols used to achieve this:

- RIP
- OSPF
- IS-IS
- IGRP/EIGRP

RIP: The Routing Information Protocol, or RIP, as it is more commonly called, is one of the most enduring of all routing protocols. RIP has four basic components: routing update process, RIP routing metrics, routing stability, and routing timers. Devices that support RIP send routing-update messages at regular intervals and when the network topology changes. These RIP packets include information about the networks that the devices can reach, as well as the number of routers or gateways that a packet must travel through to reach the destination address. RIP generates more traffic than OSPF, but is easier to configure.

RIP is a distance-vector routing protocol that uses hop count as the metric for path selection. When RIP is enabled on an interface, the interface exchanges RIP broadcasts with neighboring devices to dynamically learn about and advertise routes.

The ASA supports both RIP Version 1 and RIP Version 2. RIP Version 1 does not send the subnet mask with the routing update. RIP Version 2 sends the subnet mask with the routing

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update and supports variable-length subnet masks. Additionally, RIP Version 2 supports neighbor authentication when routing updates are exchanged. This authentication ensures that the ASA receives reliable routing information from a trusted source.

RIP has advantages over static routes because the initial configuration is simple, and you do not need to update the configuration when the topology changes. The disadvantage to RIP is that there is more network and processing overhead than in static routing.

Routing Update Process

RIP sends routing-update messages at regular intervals and when the network topology changes. When a router receives a routing update that includes changes to an entry, it updates its routing table to reflect the new route. The metric value for the path is increased by 1, and the sender is indicated as the next hop.

RIP routers maintain only the best route (the route with the lowest metric value) to a destination. After updating its routing table, the router immediately begins transmitting routing updates to inform other network routers of the change. These updates are sent independently of the regularly scheduled updates that RIP routers send.

RIP Stability Features

RIP prevents routing loops from continuing indefinitely by implementing a limit on the number of hops allowed in a path from the source to a destination. The maximum number of hops in a path is 15. If a router receives a routing update that contains a new or changed entry, and if increasing the metric value by 1 causes the metric to be infinity (that is, 16), the network destination is considered unreachable. The downside of this stability feature is that it limits the maximum diameter of a RIP network to less than 16 hops. RIP includes a number of other stability features that are common to many routing protocols. These features are designed to provide stability despite potentially rapid changes in network topology. For example, RIP implements the split horizon and hold-down mechanisms to prevent incorrect routing information from being propagated.

RIP Timers

RIP uses numerous timers to regulate its performance. These include a routing-update timer, a route-timeout timer, and a route-flush timer. The routing-update timer clocks the interval between periodic routing updates. Generally, it is set to 30 seconds, with a small random amount of time added whenever the timer is reset. This is done to help prevent congestion, which could result from all routers simultaneously attempting to update their neighbors. Each routing table entry has a route-timeout timer associated with it. When the route-timeout timer expires, the route is marked invalid but is retained in the table until the route-flush timer expires.

Procedure:

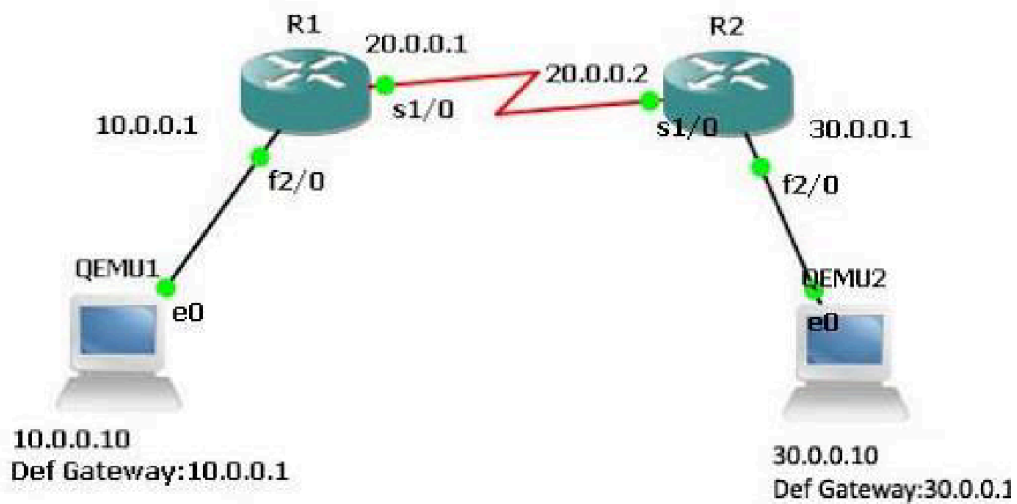
1. To begin with the experiment, open Cisco Packet Tracer
2. Create the lab and configure the basic IP addressing for all PCs and Routers
3. Using the command line interface of the router or switch enter the global configuration mode of router using command **configure terminal**
4. Steps for Configuration of RIP:

	Command	Purpose
Step 1	router rip	Starts the RIP routing process and places you in router configuration mode.
T	Example: hostname(config)# router rip	
Step 2	network network_address	Specifies the interfaces that will participate in the RIP routing process.
	Example:	If an interface belongs to a network defined by this command, the

5. Using **ping** command test the ability of the source computer to reach the destination computer

Example:

Configure RIP for following topology



COM3 - PuTTY

```
Rack2_R1(config-if)#exit
Rack2_R1(config)#interface 0/0/1
^
% Invalid input detected at '^' marker.

Rack2_R1(config)#interface gig 0/0/1
Rack2_R1(config-if)#ip address 10.0.0.1
% Incomplete command.

Rack2_R1(config-if)#ip address 10.0.0.1 255.255.255.0
Rack2_R1(config-if)#no shutdown
Rack2_R1(config-if)#ex
*Sep 30 10:40:17.468: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
Rack2_R1(config)#exit
Rack2_R1#
*Sep 30 10:40:23.523: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Sep 30 10:40:23.613: %SYS-5-CONFIG I: Configured from console by admin on console
*Sep 30 10:40:24.523: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
*Sep 30 10:42:47.531: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Sep 30 10:42:48.531: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
Rack2_R1#do show ip interface brief
^
% Invalid input detected at '^' marker.

Rack2_R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Rack2_R1(config)#do show config terminal brief
^
% Invalid input detected at '^' marker.

Rack2_R1(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  20.0.0.1        YES manual up          up
GigabitEthernet0/0/1  10.0.0.1        YES manual up          up
GigabitEthernet0/1/0  unassigned      YES NVRAM  administratively down down
Serial0/2/0          unassigned      YES NVRAM  down         down
Serial0/2/1          unassigned      YES NVRAM  down         down
Rack2_R1(config)#ip route 40.0.0.0 255.255.255.0 20.0.0.2
Rack2_R1(config)#ip route 30.0.0.0 255.255.255.0 20.0.0.2
Rack2_R1(config)#ip route 50.0.0.0 255.255.255.0 20.0.0.2
Rack2_R1(config)#
*Sep 30 11:02:45.601: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
*Sep 30 11:02:46.601: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to down
*Sep 30 11:02:52.601: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Sep 30 11:02:53.601: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

C:\ Command Prompt

```
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\CELAB>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\CELAB>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\CELAB>
```

```
C:\Users\CELAB>
C:\Users\CELAB>50.0.0.10
'50.0.0.10' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\CELAB>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:
Reply from 50.0.0.10: bytes=32 time=1ms TTL=126
Reply from 50.0.0.10: bytes=32 time=1ms TTL=126
Reply from 50.0.0.10: bytes=32 time=1ms TTL=126
Reply from 50.0.0.10: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\CELAB>_
```

CONCLUSION

In this experiment, I successfully configured the Routing Information Protocol (RIP) on the given network topology using Cisco Packet Tracer. The routers dynamically exchanged routing information and learned the routes to the remote networks. The routing table was verified using the show ip route command, and connectivity between the source and destination PCs was successfully tested using the ping command. Thus, the working of dynamic routing using RIP was studied and verified.

Signature of faculty in-charge